Proceedings of the International Conference on Internet of Things & Engineering · Published 2022

Energy-Harvesting Routing Protocols for Ultra-Dense Wireless Sensor Micro-Grids

Feng-Hua Lin, Dr. Ramesh Kulkarni, Amara Diallo

Department of Computer Science & Intelligent Systems · Publication Year: 2022

ABSTRACT

Battery-powered wireless sensor nodes impose unsustainable maintenance costs when deployed in ultra-dense agricultural and structural deployments. We introduce EH-Mesh, an opportunistic routing protocol designed for battery-less sensor nodes powered solely by ambient photovoltaic and RF energy harvesting. EH-Mesh incorporates energy-state predictive queuing, dynamically choosing forwarding paths based on anticipated ambient energy recharge rates. Simulation and hardware testbeds confirm a 99.4% packet delivery ratio and perpetual network uptime under fluctuating diurnal solar cycles. The primary limitation is high routing latency during prolonged darkness periods when nodes must enter deep-sleep states until sufficient energy accumulates.

Keywords— Internet of Things, Energy Harvesting, Wireless Routing, Ultra-Dense Networks

INTRODUCTION

Replacing thousands of chemical batteries in environmental monitoring grids is economically and ecologically unfeasible. Energy-harvesting nodes must operate intermittently, surviving power outages without losing in-flight sensor packets.

METHODOLOGY

EH-Mesh introduces capacitor voltage tracking into routing metrics. Forwarding nodes are selected based on a joint function of link quality and predicted solar irradiance using a lightweight Markov energy model.

RESULTS

Testing on 50 outdoor solar-powered sensor motes confirmed continuous operation across four seasons without battery failure, maintaining 99.4% packet delivery across 5-hop mesh trajectories.

LIMITATIONS

During overcast winter days, packet latency increased from 80ms to over 12 seconds as nodes entered intermittent sleep-and-harvest cycles while waiting for capacitor recharge.

REFERENCES

[1] Dr. Ramesh Kulkarni, Matteo Rossi, Feng-Hua Lin (2021). "Decentralized Edge Intelligence in Resource-Constrained Industrial IoT Sensor Grids". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[3] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[4] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.